PRAJWAL A

Python Developer | Generative AI & LLM Applications | Computer Vision
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SUMMARY

ECE graduate (2026) and Python developer who builds AI systems end to end: an **offline LLM learning assistant** running on Ollama, an **on-device face-recognition** pipeline on Raspberry Pi, and two **live production platforms** delivered during a 10-month internship. Looking for entry-level Python, GenAI / LLM application and AI engineering roles.

EXPERIENCE

Full Stack Developer Intern | *PulseWriteX Solutions Pvt. Ltd., Bengaluru (Hybrid)* Feb 2026 – Present

- Sole developer of **PulseX** (pulsewritexsolutions.com), a live ordering platform for PCB fabrication, 3D printing and laser cutting that replaced a fully offline quotation and order process.
- Built the backend end to end: REST APIs in **Node.js/Express**, a **MySQL** schema with **Prisma ORM**, Google OAuth and email login, and **Razorpay** payments (UPI, net banking, cards).
- Hardened a second live platform (**Rizzo Education**, AI learning): clustered deployment with **PM2 behind Nginx**, versioned database migrations, automated backups, rollback procedures and a gated **GitHub Actions CI/CD** pipeline.
- Sized and set up the production VPS from a 100–200 user load estimate; monitor logs and fix production bugs on Linux; **100+ production commits** across both platforms.

PROJECTS

Omni Room Management System – AI Smart Classroom | *Capstone* Nov 2025 – Mar 2026

Python · Ollama (local LLMs) · OpenCV · dlib / face_recognition · Raspberry Pi 4 · React · Node.js

- Built **Echo Minds**, a **local LLM** module on **Ollama** that turns lecture audio, PDFs and slide decks into **summaries, quizzes, flashcards and mind maps**, running fully offline so student data never leaves the institution.
- Benchmarked **5 face-recognition approaches** (Haar cascade, LBPH, k-NN, HOG, CNN) and shipped a **HOG detection + CNN 128-D embedding** pipeline for automatic attendance from live video.
- Integrated a custom PTZ camera (Pi 4, Sony IMX sensor, servos) streaming over HTTP into the ML backend; web platform with **4 user roles**, validated in a live demo on real classroom data.

Fluid AI – Autonomous Document Agent | *Python · FastAPI · Groq · Pydantic* 2026

- **Agentic pipeline** (validate → plan → generate → build) that turns a natural-language request into a formatted **Word document** via python-docx.
- Pydantic schemas validate every LLM response; a **retry loop feeds the exact validation error back to the model**, so malformed JSON is self-corrected instead of crashing.

Smart Attendance System – On-Device Face Recognition | *Python · OpenCV · Flask*

- Ran the **full inference pipeline on a Raspberry Pi** with no cloud: capture, **CLAHE** lighting normalisation, face detection, 128-D embedding and matching; CSV/Excel attendance export and a live **Flask** MJPEG preview.

PiPilot – Raspberry Pi Voice Assistant | *Python · Linux* In progress

- Modular **Python** assistant with **speech-to-text, text-to-speech** and command handling; config file, logging, basic tests, runs as a **systemd** service.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C
GenAI & ML: Ollama (local LLMs), Groq API, agentic pipelines, FastAPI, Pydantic, prompt design for structured outputs, OpenCV, dlib / face_recognition, HOG & CNN face models, LSTM (academic)
Backend & Data: Flask, Node.js, Express, REST APIs, MySQL, PostgreSQL, Prisma ORM
Tools & Deployment: Linux, Git / GitHub, GitHub Actions, Nginx, PM2, SSH, systemd, Raspberry Pi

EDUCATION

B.Tech, Electronics and Communication Engineering | *Dayananda Sagar University, Bengaluru* 2022 – 2026